Cognitive Fusion of Graph Neural Network and Convolutional Neural Network for Enhanced Hyperspectral Target Detection

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***Abstract*— In recent years, deep learning has emerged as a prominent technique in hyperspectral target detection (HTD). Extensive research has highlighted the potential of graph neural network (GNN) as a promising framework for exploring non-Euclidean dependencies within hyperspectral imagery (HSI). However, GNN has not been introduced to HTD. Additionally, achieving a balanced training set while effectively suppressing background remains a challenge. Therefore, we propose the cognitive fusion of GNN and convolutional neural network (CNN) for enhanced HTD (named as CFGC), which marks the first integration of GNN and CNN in HTD. Initially, using sparse subspace clustering (SSC) and a similarity measurement strategy, we select the most representative background samples for HTD. Subsequently, linear interpolation combines the prior target with the Laplacian-weighted prior target, yielding abundant targets with meaningful transformations. Finally, a fused network of CNN and GNN is utilized for training both the prior target and the constructed training set. Significantly, the incorporation of attention mechanism in both the CNN and GNN branches stands out as a noteworthy advantage, augmenting the models' ability to selectively prioritize crucial information. Four benchmark hyperspectral images have been used in extensive experiments, and the results demonstrate that CFGC exhibits superior performance in HTD.**

***Index Terms*— Attention mechanism, deep learning (DL), graph neural network (GNN), hyperspectral target detection (HTD), sparse subspace clustering (SSC).**

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I. Introduction

HYPERSPECTRAL imagery (HSI) with high spatial and spectral resolution is a 3-D image that simultaneously acquires spectral information in multiple narrow bands [1], [2], [3]. Therefore, HSI is widely used for tasks like ground object classification [4], [5], [6], change detection [7], [8], anomaly detection [9], [10], [11], [12] target detection [13], [14], [15] spectral unmixing [16], [17], and so on. Hyperspectral target detection (HTD), which identifies specific objects by extracting unique spectral features from given data with a prior target, is currently receiving increasing attention [18], [19]. Due to its ability to provide fine discrimination and detailed analysis, the task finds applications in various fields, including environmental monitoring [20], agriculture [21], geological exploration [22], military reconnaissance [23], and so on.

In HTD tasks, datasets are conventionally partitioned into target and background samples. HTD fundamentally constitutes a binary classification task [24]. Target samples, representing the objects of interest, are frequently sparsely distributed within the background samples, constituting a fraction even less than 1% of the total dataset [25].

In early HTD analysis, match filter (MF) [26] is a classical target detection method, which convolves HSI with a target spectrum to maximize correlation. The adaptive cosine estimator (ACE) [27] and the adaptive MF (AMF) [28] enhance performance by introducing adaptive features based on MF. Spectral MF (SMF) [29] also utilizes spectral information. Spectral angle mapping (SAM) [30] evaluates the similarity between spectra based on the angles to handle targets with significant spectral variations. The generalized likelihood ratio test (GLRT) [31] is able to handle complex spectral data, including nonuniform and non-Gaussian distributions.

The hypothesis testing methods mentioned above may face challenges when dealing with complex data. Hence, researchers have introduced subspace methods as an alternative approach. Orthogonal subspace projection (OSP) [32], [33] and matched subspace detector (MSD) [34] leverage the special properties of data in low-dimensional subspaces for target detection. The adaptive subspace detector (ASD) [35] and the adaptive MSD (AMSD) [36] highlight adaptability and robustness in uncertain environments. The sparse

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representation target detector (SRTD) [37], combined sparse and collaborative representation (CSCR) [38], and the prior target signature optimization-based target detector (PTSOTD) [39] find sparse representations of targets for more accurate detection. Building on the foundation, both the HTD method based on sparse representation and Cauchy distance combined graph (SRCG) [40] and the Cauchy distance graph and local adaptive collaborative representation detector (CGCRD) [14] use the Cauchy distance graph to capture the complex spatial distribution inherent in the data. Furthermore, another approach presented in [15] introduces a spectral perception and spatial–temporal tensor (SPSTT) decomposition to address similar challenges.

Constrained modeling methods in HTD involve incorporating specific constraints or prior information to enhance the accuracy and reliability of the detection process. Linear mixing models (LMMs) [41], for instance, assume that each pixel's spectrum is a linear combination of spectral signatures of different materials present in the scene. Constrained energy minimization (CEM) [42] minimizes the energy of the background signature by imposing constraints while preserving the target signature's energy. Ren and Chang [43] design target-constrained interference-minimized filter (TCIMF) which combines CEM and OSP. Ensemble-based CEM (E-CEM) [44], sparse-spatial CEM (SSCEM) [45], deep CEM (DCEM) [46], and Bayesian CEM (B-CEM) [47] are all improved algorithms of CEM, which further enhance the detection performance. Chen et al. [48] also propose a diverse-direction CEM (D2 CEM) detector to integrate local spatial information by considering eight neighborhoods surrounding each pixel in the HSI.

The abovementioned traditional methods often rely on prior knowledge and handcrafted features, whereas approaches based on deep learning automatically learn abstract features from the high-dimensional spectral vectors present in each pixel, leading to improved accuracy and robustness in detection [49]. In the preliminary exploratory phase, the convolutional neural network (CNN)-based target detector (CNNTD) [50] was proposed. The method expands the dataset through subtraction and creates a new dataset via relabeling. However, CNNTD does not tackle the issue of imbalanced sample sizes. This can make the model more prone to predicting classes with larger quantities, while overlooking classes with fewer instances, thereby impacting the accuracy of target detection [51]. In order to address the issue, Zhang et al. [52] proposed a deep CNN for HTD (HTD-Net). A modified autoencoder is introduced to generate target samples and a linear prediction strategy is used to select the background samples which are the most dissimilar to target signatures in HTD-Net. Despite the enhancements made in HTD-Net based on CNNTD, it still employs the spectral differences between pixel pairs as inputs for the network model. When performing pixel subtraction, there is a risk of losing vital information, especially in pixels surrounding the target. Zhu et al. [53] proposed the two-stream CNN for HTD (TSCNTD) to deal with the problem mentioned above. Instead of using spectral differences between pixel pairs, TSCNTD uses raw spectral feature pixel pairs for feature extraction. Due to the band redundancy in HSI, Gao et al. [54] proposed a multidepth and multibranch network for HTD based on band selection, termed as DLBSTD. Although TSCNTD and DLBSTD have achieved good classification accuracy in HTD, their capability to accurately simulate sample relationships remains limited.

Furthermore, generative adversarial network (GAN) [55], [56] has excellent generative power and diversity in HSI. Since HSI is typically spatiotemporally correlated, network models such as recurrent neural network (RNN) [57] and long short-term memory (LSTM) [58], [59] are used to capture temporal features and spatial relationships in the data. In addition, attention mechanism [60], [61] is also introduced to deal with the long-range dependencies and the global features in HSI. Although these algorithms have shown competitive detection results, in essence none of them have modeled the relationships between samples very well.

It is worth noting that graph neural network (GNN), a deep learning model designed for processing graph-structured data, is an excel in modeling relationships between individual samples [62]. In this context, nodes in the graph represent individual samples, and edges signify the relationships between these samples. GNN demonstrates exceptional proficiency in handling sparse data by effectively capturing intricate relationships and dependencies among data points within graph-structured datasets [63]. The existing CNN-based HTD algorithms have made significant progress. Therefore, integrating GNN with existing CNN models is a direction worth exploring.

In recent years, numerous GNN algorithms have been proposed for hyperspectral image classification [64], [65], [66], [67]. However, the incorporation of GNN models into HTD remains an unexplored area in scholarly research.

In summary, the HTD algorithm incorporating GNN faces several critical challenges that need to be addressed:

1) How to select representative background samples and construct sufficient target samples using HSI and limited prior targets.
2) How to use GNN models for target detection in HSI.
3) How to retain the advantages of the CNN model while using GNN to model the relationship between pixels in HSI.

Based on the above analysis, we propose the cognitive fusion of GNN and CNN (CFGC) algorithm, which consists of a background sample selection strategy, a target random weight representation, and a cognitive fusion network integrating GNN and CNN. In the background sample selection phase, we utilize sparse subspace clustering (SSC) to identify distinct subspaces within HSI. Each pixel is projected into its corresponding subspace, capturing its spectral characteristics accurately. Using the Euclidean distance metric, we measure the dissimilarity between the pixels, selecting the pixels farthest from the prior target spectrum as the representative background sample for each subspace. During the target generation process, the temporary targets are initially generated using Laplacian weights. Subsequently, the temporary targets are linearly interpolated with the prior target to obtain the final targets. Each target sample retains the essential traits of the

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prior target while introducing variability through these random feature weights. The network model is constructed through the integration of two branches: the CNN branch and the GNN branch. The training dataset and the prior target are fed separately into the upper and lower branches of the CNN to extract features by dilated convolution. Meanwhile, the GNN branch consists of graphSAGE [68] and graph transformer [69]. The fusion features are obtained by elementwise addition. During the testing, the HSI and the prior target are fed into the trained CFGC. In summary, the innovations of the proposed CFGC are given as follows:

1) A novel method for constructing the background training set is introduced, enhancing the accuracy of subsequent analysis. Initially, the pixels of HSI are categorized into multiple classes using the SSC algorithm. Subsequently, the subspace projection method is employed to select the most representative background samples. Compared to other background sample selection approaches, the background samples selected by the proposed method are more accurate and reliable.
2) Aiming at the serious training sample imbalance of CFGC, a new approach for generating target samples is proposed. The target sample is obtained through linear interpolation blending the prior target with the prior target after Laplacian weighting. The method avoids introducing too much noise while achieving target diversity.
3) GNN is introduced for the first time in HTD, leading to the creation of a specialized GNN branch within the detection network. In this branch, local node information is first aggregated using graphSAGE. Then a graph transformer is used to model the distance-dependent relationships between nodes and their neighborhood information. This approach enables a comprehensive analysis of HSI by effectively capturing both local and global spectral dependencies, resulting in the suppression effect on background samples.
4) A cognitive fusion of GNN and CNN is achieved, resulting in a novel HTD network. The CNN branch focuses on extracting localized spatial information. In contrast, the GNN branch specializes in capturing medium- to long-range spatial relationships among samples. By fusing these two branches, the model effectively integrates both local and mid-to long-range spatial information, enhancing its ability to analyze complex spatial patterns in HSI.

The subsequent sections of this article are structured as follows: In Section II, we provide a concise overview of the pertinent knowledge. Section III elaborates on the comprehensive pipeline of the proposed CFGC. Subsequently, in Section IV, we substantiate the efficacy and superiority of CFGC through extensive experiments and in-depth analysis. Finally, the findings of this study are summarized in Section V.

II. Preliminary and Related Work

A. Problem Definition of HTD Based on GNN

GNN excels in achieving precise target detection by thoroughly exploring the interconnections within the pixels. For easy understanding, HSI is assumed to be represented as $\boldsymbol{I} \in R^{H\times W\times C}$, where H, W, and C denote the height, width, and number of spectral bands of data cube, respectively. In essence, the objective of HTD based on GNN is to categorize each pixel within a hyperspectral image, determining whether it belongs to a specific target. The spectral relationship can be represented as an undirected graph. Let $G = (V, E)$ be an undirected graph, where V represents the vertex and E represents the edge sets. The vertex set comprises $\boldsymbol{N} = H \times W$ pixels, with the edge set representing the connections between any two vertices, denoted as v_{i} and v_{j}. Therefore, $\boldsymbol{I}$ can be converted into the feature matrix $\boldsymbol{X} \in R^{N\times C}$. The adjacency matrix represents the connection patterns between nodes within a graph structure, which is denoted as $\boldsymbol{A} \in R^{N\times N}$. The calculation method of the adjacency matrix is depicted in the following equation:

$$\boldsymbol{A}_{i,j} = \begin{cases} 0, & \text{if } v_i \text{ and } v_j \text{ is adjacent} \\ 1, & \text{otherwise.} \end{cases} \tag{1}$$

Therefore, the degree of each node in the graph is obtained by summing the elements in each row of the adjacency matrix, as shown in the following equation:

$$\boldsymbol{d}_i = \sum_{j=1}^{N} \boldsymbol{A}_{ij}. \tag{2}$$

Thus, the degree matrix $\boldsymbol{D} = \mathrm{diag}(d_1, d_2, \ldots, d_{\mathrm{N}})$ is also obtained. In the HTD task, pixels are categorized into positive and negative samples, making HTD a binary classification problem. Node classification based on GNN involves representing each node in the graph as a vector and employing an iterative feature propagation process. During this iterative process, the node continuously aggregates information from its neighbors. The feature propagation paradigm of GNN in the kth iteration is represented by the following equation:

$$\boldsymbol{h}_{\boldsymbol{v}_i}^{(k+1)} = \mathrm{UPDATE}^{(k)}\left(\boldsymbol{h}_{\boldsymbol{v}_i}^{(k)}, \boldsymbol{m}_{\boldsymbol{N}(\boldsymbol{v}_i)}^{(k)}\right) \tag{3}$$

where $\boldsymbol{m}_{\boldsymbol{N}(v)}^{(k)} = \mathrm{AGGREGATE}^{(k)}(\{\boldsymbol{h}_u^{(k)} | \forall u \in \boldsymbol{N}(v)\})$. The hidden embedding of vertex $\boldsymbol{v}_{\mathrm{i}}$, denoted as $\boldsymbol{h}_{\boldsymbol{v}_i}^{(k+1)}$, can be updated by aggregating features of $\boldsymbol{N}(v)$, which is the neighborhood of $\boldsymbol{v}_{\mathrm{i}}$. Assuming that propagation is completed after the kth iteration, the model has learned the final feature representations for each node $\boldsymbol{v}_{\mathrm{i}}$, which is denoted as $\boldsymbol{H}_{\boldsymbol{v}_i} = \boldsymbol{h}_{\boldsymbol{v}_i}^{\boldsymbol{K}}$. Finally, an activation function is employed to predict the probability of a pixel being the target. Its expression is defined in the following equation:

$$\boldsymbol{Y}^* = \mathrm{Sigmoid}(\boldsymbol{f}(\boldsymbol{H}_{\boldsymbol{v}_i})) \tag{4}$$

where the $\boldsymbol{f}(\bullet)$ is the fully connected layer, and the Sigmoid activation function is finally employed to acquire the final decision vector $\boldsymbol{Y}^*$.

B. Sparse Subspace Clustering

Consistent with Section II, HSI is represented as $\boldsymbol{I} \in R^{H\times W\times C}$ $(N = H \times W)$. The SSC method utilizes the low-dimensional sparse characteristics inherent in HSI by

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mapping the data into a lower-dimensional subspace. Consequently, the method adeptly handles HSI's high dimensionality and identifies clear features to cluster equivalent pixels. Each pixel within HSI dataset is considered an independent sample originating from a union of subspaces $\{U_i\}_{i=1}^{l} S_i$, where each subspace corresponds to a distinct object category. The SSC algorithm, driven by the intrinsic self-representation property of the data, employs a judicious combination of linear or affine relationships with a minimal set of data points to effectively represent each pixel within the union of these subspaces [70]. The affiliations of data points with the underlying subspaces are recorded in the sparse coefficient matrix, denoted as $\boldsymbol{Z} \in R^{\mathrm{N}\times\mathrm{N}}$, and are optimized through the subsequent objective function. The fundamental concept lies in the optimization objective, as shown in the following equation:

$$\min_{\boldsymbol{Z}} ||\boldsymbol{Z}||_1 + \frac{\lambda}{2}||\boldsymbol{Y} - \boldsymbol{Y}\boldsymbol{Z}||_F^2 \quad \text{s.t. } \operatorname{diag}(\boldsymbol{Z}) = 0, \quad 1^T\boldsymbol{Z} = 1^T \tag{5}$$

where $\boldsymbol{Y} \in R^{C\times N}$ represents the 2-D matrix of his, and λ represents the parameter which regulates the regularization terms. In addition, the l_1-norm is employed to encourage sparsity by promoting the presence of a small number of significant elements in a vector, whereas the F-norm is utilized to promote matrix denseness, ensuring a more uniform distribution of elements across the matrix. The constraint $\operatorname{diag}(\boldsymbol{Z}) = 0$ means that the diagonal elements of the matrix $\mathbf{C}$ are equal to zero. In a sparse representation, the constraint promotes sparsity by encouraging the usage of fewer nonzero coefficients to represent the data effectively. The constraint $1^T\boldsymbol{Z} = 1^T$ ensures that the sum of coefficients in each column of the sparse coefficient matrix $\boldsymbol{Z}$ equals 1. In this context, it guarantees that each data point is represented as a sparse linear combination of the other data points, facilitating effective data representation and clustering.

Utilizing the alternating direction method of multipliers (ADMMs) [71], solving the SSC model and acquiring the sparse coefficient matrix $\boldsymbol{Z}$ become straightforward. Subsequently, the construction of the similarity matrix $\boldsymbol{S}$ ensues, as shown in the following equation:

$$\boldsymbol{S} = |\boldsymbol{Z}| + |\boldsymbol{Z}|^T. \tag{6}$$

The ultimate clustering outcome can be derived through the application of the spectral clustering algorithm to the similarity matrix.

III. Proposed Method

In this section, a detailed description of the proposed method CFGC is provided. The workflow of CFGC, depicted in Fig. 1, comprises two primary stages: the training stage and the testing stage. During the training stage, a series of steps are undertaken, including the selection of background samples, generation of target samples, and training of the fusion network comprising both GNN and CNN. During the testing stage, detected pixels are combined with target priors, creating the testing samples. These samples are subsequently fed into the well-trained fusion network for classification. As a result, the proposed CFGC method generates the ultimate detection outcomes. In addition, the details of CFGC are shown in Fig. 2.

A. Background Sample Selection

To ensure accurate characterization of background samples in HTD, we introduced an approach combining SSC with a similarity measurement strategy for the selection of background samples. Fig. 1 depicts a flowchart outlining the process of background sample selection.

In tackling the challenge posed by the curse of dimensionality in HSI, we initially applied principal component analysis (PCA) to HSI. Subsequently, SSC was employed which delineates data points into m clusters where each cluster corresponds to a low-dimensional subspace. In the ith cluster, there are m_i pixels, which can be represented as follows:

$$\boldsymbol{S}_i = \{\boldsymbol{s}_{i1}, \boldsymbol{s}_{i2}, \ldots, \boldsymbol{s}_{im_i}\}, \quad 1 \le i \le m. \tag{7}$$

Next, we further select the most representative background samples based on the clustering results. This selection process hinges on the Euclidean distance. The Euclidean distance, simple and practical, quantifies spatial distances between samples. Distant samples, often with larger differences in feature space, are more likely to represent background samples. Precisely, we choose pixels that differ the most from the prior target, meaning they are the farthest away from the prior target in terms of Euclidean distance. In the ith cluster, the selected background sample is shown in the following equation:

$$\boldsymbol{b}_i = \arg\max_{\boldsymbol{s}_i \in \boldsymbol{S}_i} ||\boldsymbol{s}_i - \boldsymbol{p}||_2^2 \tag{8}$$

where p represents the prior target. We use the square of the Euclidean distance for similarity measure to highlight samples far away from the target, which helps to select more representative background samples. To prevent overfitting, the number of clustered categories is restricted. Therefore, we select a certain proportion of background samples in each category. We use B to represent the selected background sample set, as shown in the following equation:

$$B = \sum_{i=1}^{m} \boldsymbol{b}_i. \tag{9}$$

The number of background samples selected in each cluster is determined by the ratio of the number of samples in the cluster to the total number of samples.

Compared to conventional clustering algorithms, SSC's robustness to noise and outliers, along with its ability to handle nonconvex clusters, enables accurate identification of the diverse backgrounds materials commonly encountered in HTD. Further applying a similarity measurement strategy on the clustered results to select samples that are least similar to the prior target can effectively suppress background interference. In conclusion, this meticulous methodology guarantees the stringent selection of representative background samples, consequently improving the dependability and precision of HTD.

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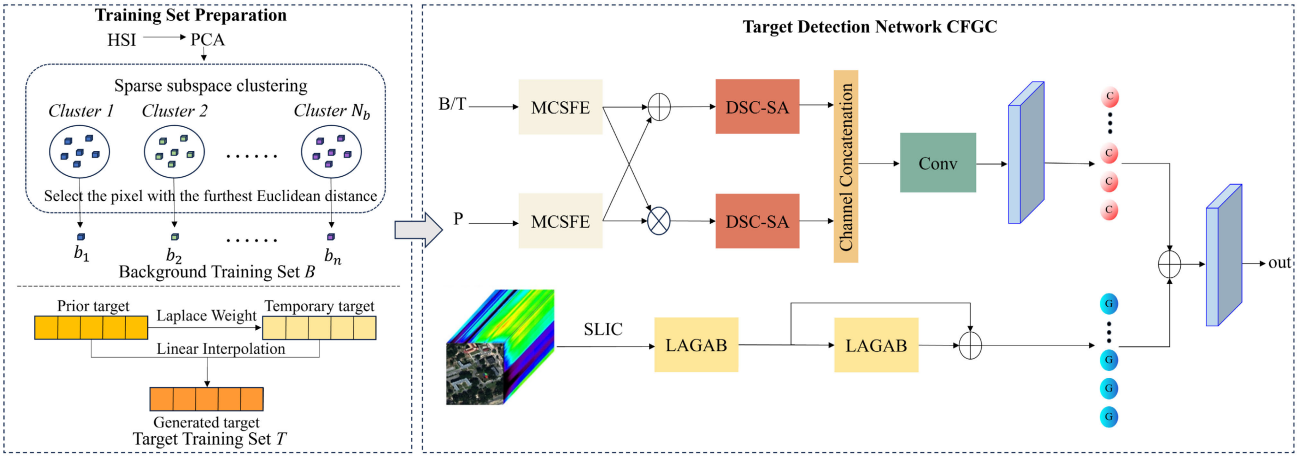


Fig. 1. Framework of the proposed method. Here, the details of CFGC are shown in Fig. 2.

Algorithm 1 Proposed CFGC

Input: The hyperspectral image and the prior target.

1. Utilizing the SSC algorithm and the similarity metric strategy to select background samples.
2. Constructing target samples through the proposed LapMix method.
3. Generating training samples: positive samples (labeled as 1) are formed by using the prior target and target samples, and negative samples (labeled as 0) are constructed through the prior target and background samples.
4. Training the proposed CFGC network using the training samples.
5. Inputting the test samples into the trained network.
6. Obtaining the output for each test pixel to form the final detection results.

Output: The detection results of the proposed CFGC.

Algorithm 2 LapMix

Input: $\boldsymbol{p}$-prior target, N_t-number of the target training samples.

Output: T-the target training set.

1. Initialize target training set $T = \emptyset$;
   Initialize mixup parameter λ with a random value from a uniform distribution [1, 0];
2. for $i = 1$: N_t do
3. Initialize the i-th target training sample $\boldsymbol{t}_i = \boldsymbol{p}$;
4. Initialize the temporary target sample $\boldsymbol{t}_{temp} = \boldsymbol{p}$;
5. Generate a Laplace-distributed weight $\boldsymbol{w}$ with the same dimensions as $\boldsymbol{p}$;
6. $\boldsymbol{t}_{temp} = \boldsymbol{p} \odot \boldsymbol{w}$;
7. $\boldsymbol{t}_i = \lambda * \boldsymbol{p} + (1 - \lambda) * \boldsymbol{t}_{temp}$;
8. $T = T \cup \boldsymbol{t}_i$;
9. end

B. Target Sample Generation

Inspired by the principles of Laplace weighting and mixup [72], we propose an effective target generation method named LapMix. LapMix relies solely on the prior target, avoiding the dilution of target features with irrelevant background information.

As shown in Fig. 3, the original mixup blends input data by linearly interpolating features and labels of two distinct samples, creating a new synthetic training example. Specifically, given feature tensors $\boldsymbol{X}_i$ and $\boldsymbol{X}_j$ corresponding to two samples along with their respective labels Y_i and Y_j, mixup generates a new sample as indicated by (10), with the corresponding label being generated according to (11)

$$\tilde{\boldsymbol{X}} = \lambda \boldsymbol{X}_i + (1 - \lambda)\boldsymbol{X}_j \tag{10}$$

$$\tilde{Y} = \lambda Y_i + (1 - \lambda)Y_j. \tag{11}$$

Here, λ is a parameter randomly sampled from a Beta distribution, typically ranging between 0 and 1.

LapMix generates Laplace weight with the same dimension as the prior target. Subsequently, the elementwise product of the weight and the prior target yields temporary target sample. Linear interpolation is then applied to combine the features of the temporary target and prior target, resulting in individual target sample. This process is iteratively repeated to construct the target training set. The specific process of LapMix is illustrated in Algorithm 2.

Incorporating Laplace weight into the generation of target samples for HTD enables accurate predictions by effectively capturing the uncertainties and distribution characteristics inherent in the data. The interpolation fusion integrates the information of the prior target, introducing target diversity without excessive deviation. Unlike traditional mixup, LapMix selectively combines only sample features. The shared label between temporary target and prior target eliminates the need for label fusion. This strategic blending of samples with identical labels results in a more diverse array of data samples, alleviating the scarcity of target samples in HSI.

C. GNN and CNN Fusion Network

To effectively integrate the local and mid- to long-range spatial information of HSI, we introduce the CFGC algorithm. The specific workflow of CFGC is illustrated in Fig. 1, comprising the CNN branch, GNN branch, and fusion operation.

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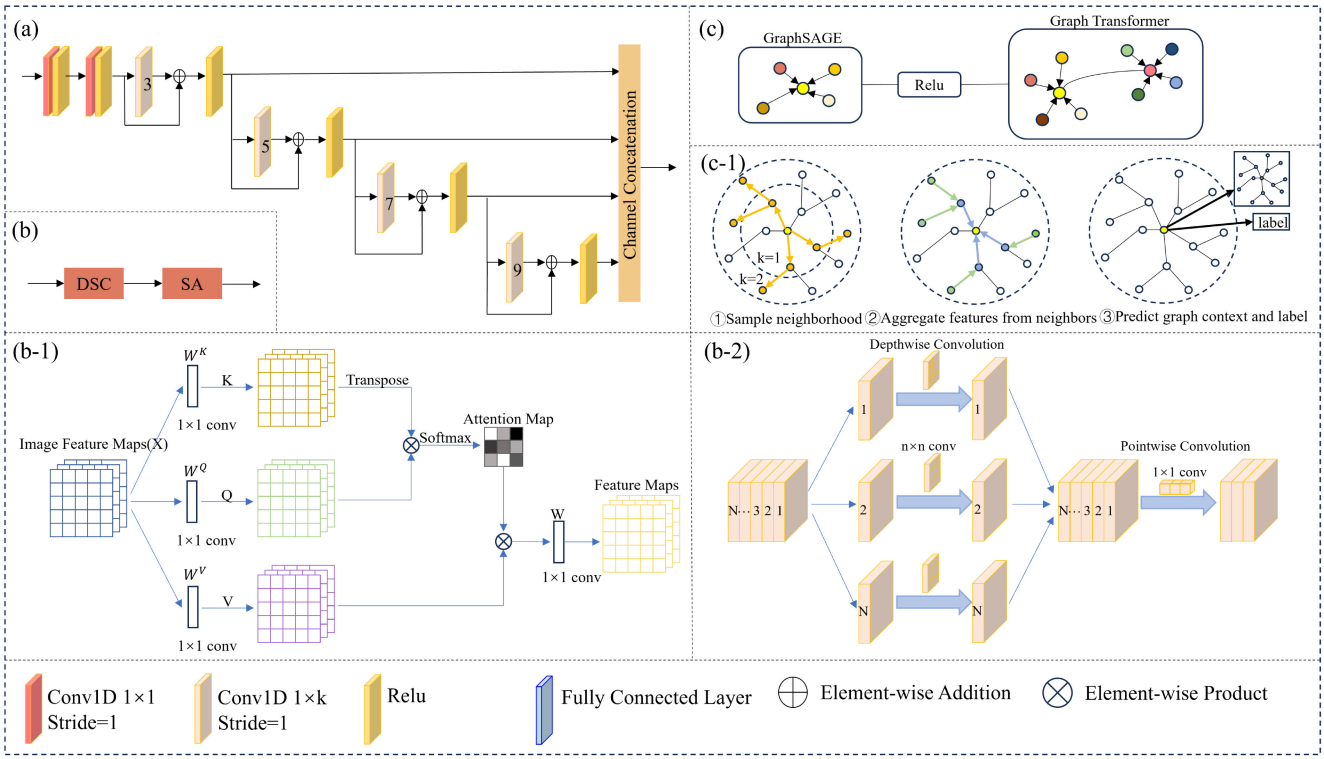


Fig. 2. Implementation details of CFGC. (a) MCSEF in CFGC. (b) DSC-SA in CFGC. (b-1) SA in DSC-SA. (b-2) DSC in DSC-SA. (c) LAGAB in CFGC. (c-1) GraphSAGE in LAGAB.

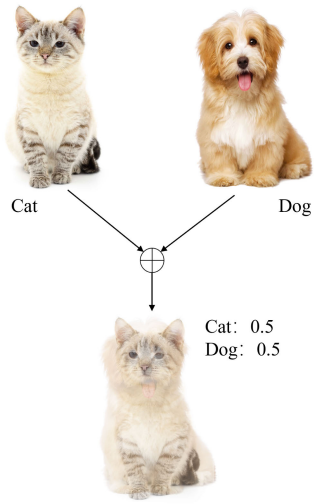


Fig. 3. Framework of mixup.

The concrete architecture of the CNN branch is delineated in Fig. 1, comprising principally the multiscale channel spectrum feature extractor (MCSFE) module, the depthwise separable convolution-self attention (DSC-SA) module, and the Conv block. The MCSFE module employs convolutional kernels of varying sizes to capture diverse scale information within the input data. Furthermore, feature fusion is achieved by concatenating the channels of convolutional output at each layer. The DSC-SA module combines the features of depthwise separable convolution (DSC) block and self-attention (SA) block. The DSC block is utilized for extracting deep contextual information, while the SA block enhances important scale features. Particularly, the DSC block is shown in Fig. 2(b-2), which decomposes the standard convolution operation into depthwise and pointwise convolutions. It first performs depthwise convolutions, applying a separate convolutional filter to each input channel, followed by pointwise convolutions, which combine the spatial information. This factorization reduces computational complexity and parameters in CNN branch. The SA block is designed to enhance the capture of long-range spatial dependencies within the localized context. It calculates attention weights for each pixel based on its relationships with all other pixels, allowing the CNN branch to dynamically focus on more relevant information during processing.

The target and background training sets are fed into the upper branch of CNN, while the prior target is sent to the lower branch. Initially, through the MCSFE module, multiscale convolution features (f_p and f_d) are obtained from the two branches. Furthermore, to effectively discern the distinctions or similarities between the background/target training set and the prior target, we employ both addition and multiplication as feature fusion operations. Thus, two fusion features are obtained, namely $f_+ = f_p + f_d$, $f_\times = f_p \times f_d$. In this context, addition enhances the influence of prior information, while product not only amplifies information resembling the prior target but also augments the capacity for nonlinear expression. Later, the features f_+ and $f_\times$ are fed into the DSC-SA module, resulting in two enriched features at both spatial and contextual levels. Afterward, the two features are channel-concatenated and input into the Convolutional (Conv) block, which composed of convolutional and activation layers, generating higher-level abstract features. Finally, the abstract

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Algorithm 3 GraphSAGE Local Node Aggregation Algorithm

Input: $\boldsymbol{G} = (\boldsymbol{V}, \boldsymbol{E})$_ graph, $\{\boldsymbol{x}_v, \forall \boldsymbol{v} \in \boldsymbol{V}\}$—input features, K—the number of depth of the network, $\boldsymbol{w}^k, \forall k \in \{1, \ldots, K\}$−weight matrices, σ_ nonlinearity, MEAN—differentiable aggregator functions; N:$v \rightarrow 2^v$_ neighborhood function.

Output: Vector representation $\boldsymbol{z}_v$ for all $v \in V$.

1. $\boldsymbol{h}_v^\circ \leftarrow \boldsymbol{x}_v, \forall v \in V$;
2. for $k = \{1, \ldots, K\}$ do
3. for $v \in V$ do
4. $\boldsymbol{h}_v^k \leftarrow \sigma(\boldsymbol{w}^k \cdot MEAN(\{\boldsymbol{h}_v^{k-1}\} \cup \{\boldsymbol{h}_u^{k-1}, \forall u \in (N(v))\}))$
5. end
6. $\boldsymbol{h}_v^k \leftarrow \frac{\boldsymbol{h}_v^k}{||\boldsymbol{h}_v^k||_2}, v \in V$
7. end
8. $\boldsymbol{Z}_v \leftarrow \boldsymbol{h}_v^k, \forall v \in V$.

features undergo dimension transformation through a fully connected layer, yielding the extracted features from the CNN branch.

The LAGAB module is composed by concatenating the graphSAGE and the graph transformer. The graphSAGE embedding generation algorithm is described in Algorithm 3. In particular, $\forall \boldsymbol{x}_v$ represents the input eigenvector of node v, and K signifies the maximum number of hops for each vertex to aggregate information from its neighboring vertices. As illustrated in Fig. 2(c-1), when $K = 2$, each vertex can learn its own embedding representation based on the information from its 2-hop neighbors. In addition, $N(v)$ denotes the sampled neighbor nodes of node v, while $\boldsymbol{h}_v^k$ denotes the feature representation encompassing all neighboring nodes of node v. As illustrated in Algorithm 3, the first step is to selectively sample the neighboring nodes of the central vertex within the graph, followed by the aggregation of features from both the central vertex and its neighboring nodes. This process culminates in the creation of the embedding representation for the central vertex. Hence, the features that capture both spatial and spectral dimensions are extracted and the feature representation of the center node is also enhanced. However, the relationships of remote nodes have not been effectively modeled. The graph transformer integrates the transformer architecture into GNNs, introducing SA mechanism for enhanced relational modeling. It operates by first embedding nodes into high-dimensional vectors, then applying SA across nodes to dynamically weight their contributions based on learned relationships. This mechanism enables the model to capture long-range dependencies and global relationships within the graph. The inclusion of layer normalization ensures stable training, while feedforward networks contribute to the overall expressive power of the model. The module lies in its ability to leverage the self-attention mechanism, originally designed for sequential data, to effectively capture complex relationships in graph-structured data.

In the GNN branch, the dataset undergoes superpixel segmentation, leading to the partitioning of the dataset into superpixel blocks. These blocks are subsequently fed into two LAGAB modules. Subsequently, based on the batch processed by the CNN branch, the corresponding features for the GNN branch are selected. To be more specific, following the sequential labeling of original images, we then calculate the image coordinates within the batch using division and modulo operations. Subsequently, we reference these coordinates to determine the corresponding superpixel block number. With the superpixel block number identified for each specific pixel, we extract the features of the corresponding superpixel. Finally, we merge the features of the superpixels with the batch features obtained from CNN through a weighted addition. This fusion approach allows us to effectively leverage both CNN and GNN features, thereby enhancing performance in HTD. The combined features then pass through a fully connected layer and a sigmoid function to generate the network's output.

Then, we employ the binary cross entropy (BCE) loss to quantify the dissimilarity between predicted probabilities and actual labels. The BCE loss is represented as $\text{BCE} = -(1/M)\sum_{i=1}^{M}[y_i \cdot \log f_i + (1 - y_i) \cdot \log(1 - f_i)]$, where M represents the batch size, f_i denotes the output of the network, and y_i corresponds to the positive or negative labels, namely 1 or 0.

During the testing phase, both the pixels under evaluation, denoted as $\boldsymbol{t}_k (k = 1, 2, \ldots, N)$, and the prior target p, are simultaneously input into the pretrained CFGC network. The resulting output, denoted as $R_{\text{CFGC}}(\boldsymbol{t}_k)$, represents the probability of $\boldsymbol{t}_k$ being the target. Ultimately, the classification of the pixel $\boldsymbol{t}_k$ during testing is determined as either the target or background by applying a dynamic threshold η. If the result of the function $R_{\text{CFGC}}(\boldsymbol{t}_k)$ is greater than η, $\boldsymbol{t}_k$ is classified as the target; otherwise, it is categorized as the background.

IV. Experimental Results and Analysis

A. Hyperspectral Datasets

Four benchmark datasets are utilized to evaluate the detection capabilities of the proposed CFGC. Fig. 4 displays the pseudocolor images along with their respective ground-truth maps.

1) HYDICE Dataset: The dataset is a subset derived from the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor, situated in the urban area of California, USA. This subimage comprises 80 × 100 pixels, featuring a spatial resolution of 2 m. Additionally, the spectral bands of the dataset have been narrowed down to 162 to enhance the signal-to-noise ratio (SNR) in our experiments. The dataset focuses on targets of interest, namely cars and roofs, which collectively consist of 21 pixels.

2) San Diego Dataset: It is a publicly available HSI dataset captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) at San Diego Airport [73]. The dataset boasts 224 spectral bands and a resolution of 3.5 m. Due to considerations such as water vapor absorption and low SNR, 35 suboptimal spectral bands have been eliminated [74]. Therefore, the dimension of the data cube is 100 × 100 × 189. The target samples in the dataset consist of 64 pixels arranged in three planes.

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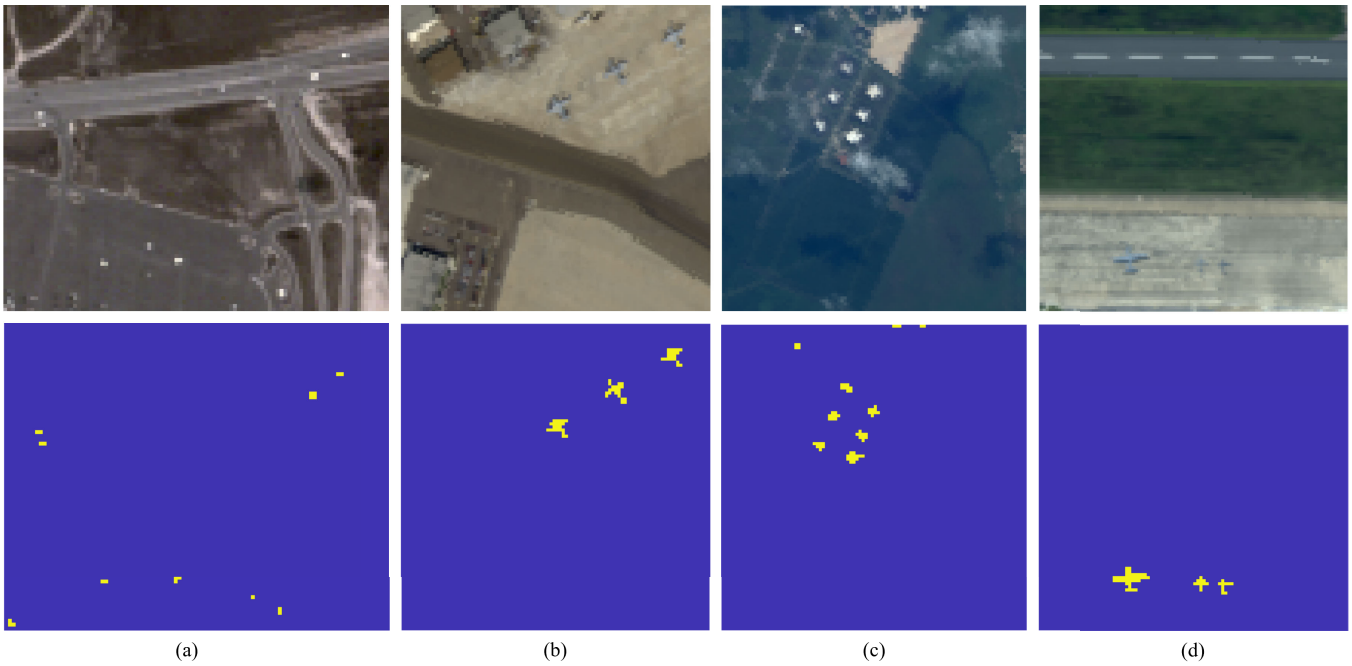


Fig. 4. Pseudocolor images and ground-truth maps of the four experimental datasets. (a) HYDICE. (b) San Diego. (c) Urban-1. (d) Airport-4.

3) Airport Beach Urban Dataset: This dataset comprises five urban images, four beach images, and four airport images. In the experiment, we selected one image each for target detection from the urban and airport images, specifically identified as Urban-1 and Airport-4, respectively. Both of the subdatasets were acquired through AVIRIS [75]. Urban-1 dataset was obtained from the Texas Coast, USA, with a scene size of 100 × 100 pixels and a spatial resolution of 17.2 m. After excluding bands affected by water absorption, 204 bands from the original dataset were retained. Airport-4 dataset comprises 100 × 100 pixels with a total of 191 spectral bands. Additionally, the targets of interest in this dataset are composed of 60 pixels.

B. Evaluation Indicators

To assess the efficacy of the CFGC, three quantitative evaluation metrics are utilized, including the receiver operating characteristic (ROC) curve, the area under the curve (AUC) value, and the separability map.

1) ROC Curve: It visually illustrates the balance between true positive rates and false positive rates across diverse threshold settings. The x-axis represents the false positive rate which means the proportion of false positives among all actual negatives. The y-axis represents the true positive rate, which means the proportion of true positives among all actual positives. A curve closer to the upper-left corner indicates superior model performance, as it achieves higher true positive rates while maintaining lower false positive rates.

2) AUC Value: The AUC value quantifies the area beneath the ROC curve, commonly computed through numerical integration methods. A heightened AUC value, approaching 1, signifies enhanced capacity in distinguishing between the target and background samples. This metric proves especially advantageous in scenarios marked by imbalanced class distribution, as it provides a holistic assessment of model performance, independent of class proportions [76].

3) Separability Map: The separability map functions as a spatial representation, visually indicating the distinguishability between different classes within HSI. This map is instrumental for identifying regions of clear separation as well as areas where target and background may overlap. The separability map is visually represented by green and orange boxes, symbolizing background and target samples, respectively. The closer the green box is to ordinate 0, the more effectively the algorithm suppresses background information. The closer the orange box is to ordinate 1, the more prominently targets are represented in the scene.

C. Experiment Settings

In order to mitigate potential errors, a statistical approach is employed by computing the mean of ten experimental trials for all detection results. Different detectors utilize the same prior targets on the same dataset. The anomaly detection dataset does not have any predefined prior targets. Instead, the prior targets are derived by randomly selecting ten target pixels and computing their averages along the spectral dimension.

In the experimental process, we configured the batch size to 512. Additionally, the learning rate serves as a pivotal hyperparameter that profoundly influences the efficiency of the optimization process during model training. When the learning rate is set excessively small, the training process is afflicted with lethargic convergence. Conversely, an overly large learning rate can lead to divergence and instability. Therefore, we set the learning rate to 0.0001, resulting in reduced training time and improved training effectiveness.

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To assess the performance of the proposed CFGC, a comparative analysis was conducted against nine state-of-the-art target detection methods. These include three classical detectors, namely SAM [30], OSP [32], and CEM [42]; two advanced feature-based detectors, namely SRTD [37] and CSCR [38]; as well as four deep learning-based approaches, namely CNNTD [50], HTD-Net [52], TSCNTD [53], and STTD [61].

Comparator detector parameters were configured to match those specified in the original text, while platform-specific parameters were fine-tuned for optimal performance. In the experimental phase, we obtained the parameters for each detector through cross-validation. Due to variations in the sizes of targets across different datasets, the sizes of the dual concentric windows involved in CSCR and SRTD are set inconsistently. In our experiments, the inner and outer window sizes are configured as follows: 11×11 and 21×21 for San Diego and Airport-4, and 5×5 and 7×7 for Urban-1 and HYDICE.

D. Parameter Analysis

1) Weight of GNN features: CNN excels at extracting local spatial information, while GNN is proficient at capturing spatial relationships between pixels at medium to long distances. Additionally, GNN effectively captures relationship between pixels in graph-structured data, leading to a certain degree of suppression effect on background samples. Therefore, it is crucial to determine the appropriate fusion weight for the features extracted from the CNN and GNN branches, as it can optimize the overall performance of the model. In light of this, we discuss the impact of different weights of the GNN feature on HTD. The relationship between the AUC values and GNN feature weight is illustrated in Fig. 5. It is observed that low GNN feature weight diminishes the model's ability to capture pixel spatial relationships, resulting in the loss of crucial contextual information and a decrease in target detection performance. Conversely, when GNN feature weight is set to larger values, a balance between local and global information is achieved, resulting in higher AUC values. However, excessively large GNN weights tend to neglect local spatial details, leading to relatively lower AUC values. Therefore, optimal performance is attained only when GNN weight is set appropriately. On the HYDICE, San Diego, and Urban-1 datasets, the weight of features extracted from the GNN branch is set to 0.4. Meanwhile, on the Airport-4 dataset, the weight is set to 0.2.

2) λ in LapMix: The mixup parameter λ in LapMix controls the relative contributions of the prior target and the temporary target during the mixing process, thereby influencing the diversity of the constructed target samples. Therefore, selecting appropriate parameters is crucial for enhancing subsequent target detection performance. In this regard, we explore the impact of different values of λ on HTD. The relationship between AUC values and λ is illustrated in Fig. 6. It is observed that when λ is set to a very small value, the constructed target samples tend to favor the temporary target, introducing more noise and resulting in imprecise handling of details and edge cases. Conversely, when the parameter is set to a larger value, diverse target samples are generated while avoiding excessive noise, making the model more robust and achieving the maximum AUC value. However, setting the parameter to an excessively large value causes the constructed target samples to lean toward the prior target, lacking sufficient richness and leading to overfitting of the original prior target, poor generalization, and a decrease in AUC value. Therefore, optimal results can only be achieved with a judicious choice of the λ. Experimental results indicate that, on the four datasets used in this article, the mixup parameter λ is set to 0.7.

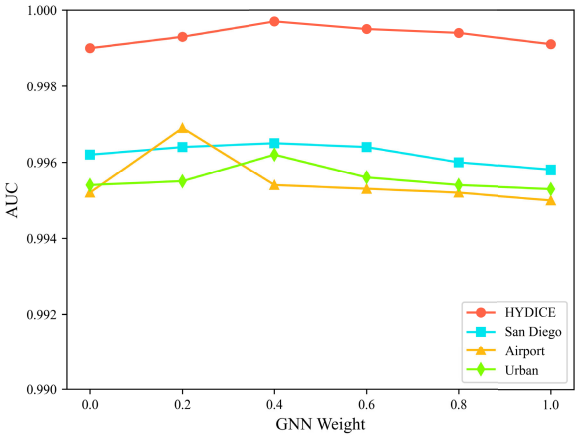


Fig. 5. Analysis of selected GNN weight.

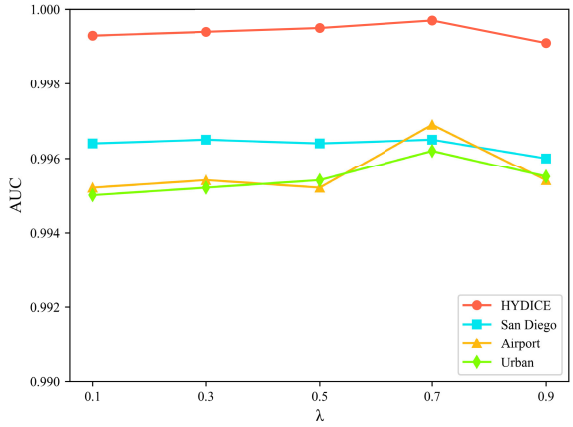


Fig. 6. Analysis of selected λ.

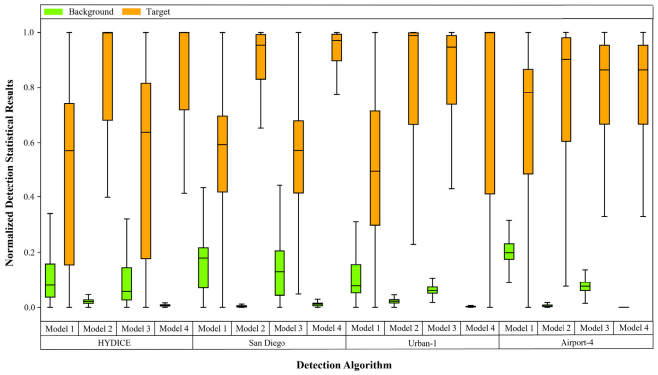


Fig. 7. Separability map of ablation study on four experimental HSI datasets.

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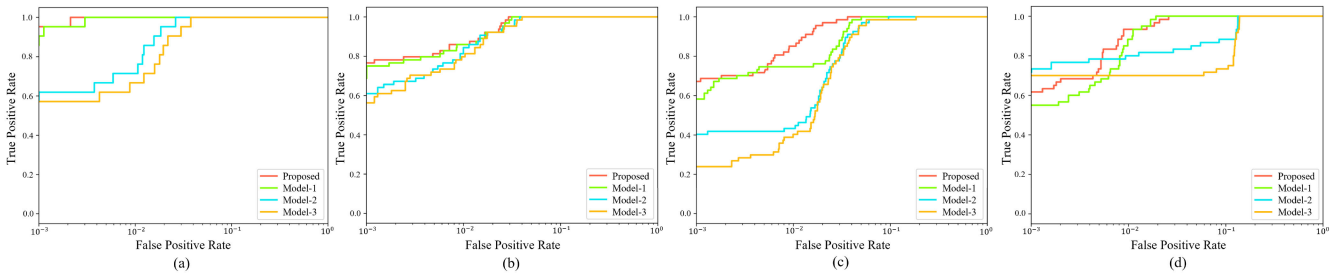


Fig. 8. ROC curves of ablation study on four experimental HSI datasets. (a) HYDICE. (b) San Diego. (c) Urban-1. (d) Airport-4.

TABLE I
AUC VALUES OF CFGC AND ITS DIVERSE VARIANTS ON FOUR BASELINE DATASETS

Datasets	Model	DSC-SA	GNN-branch	AUC
HYDICE	1	×	×	0.99199
	2	✓	×	0.99961
	3	×	✓	0.99496
	4	✓	✓	0.99973
San Diego	1	×	×	0.99459
	2	✓	×	0.99613
	3	×	✓	0.99511
	4	✓	✓	0.99657
Urban-1	1	×	×	0.98051
	2	✓	×	0.99182
	3	×	✓	0.98392
	4	✓	✓	0.99627
Airport-4	1	×	×	0.96382
	2	✓	×	0.99616
	3	×	✓	0.98062
	4	✓	✓	0.99691

Furthermore, varying parameter values consistently yield high AUC values for the corresponding detection results (all above 0.99), demonstrating the superiority of the proposed CFGC.

E. Ablation Study

To validate the effectiveness and superiority of the proposed DSC-SA module and GNN branch, we explored four variations regarding the use of these components. To ensure fairness, the structures and parameter settings of other modules were kept consistent, except for the discussed components. The model configurations and obtained AUC values for the four variants are presented in Table I. Model 1 represents the model without the DSC-SA module and GNN branch. Model 2 retains the DSC-SA module and removes the GNN branch. Model 3 removes the DSC-SA module but retains the GNN branch. Model 4 represents the CFGC model, incorporating both the DSC-SA module and GNN branch. Separability maps are depicted in Fig. 7, and ROC curves are illustrated in Fig. 8 as well.

As shown in Table I, Model 1 achieved the lowest AUC value. In comparison, Model 2, although not utilizing the GNN branch, exhibited improved AUC values across all four datasets when the DSC-SA module was added. This validates the effectiveness of the DSC-SA module. DSC extracts deep contextual information and SA reinforces important scale features, thereby enhancing the detection performance of the model. Similarly, Model 4, incorporating the DSC-SA module, outperformed Model 3 in terms of AUC values. Furthermore, comparing the results of Model 1 with Model 3 and Model 2 with Model 4 revealed an enhancement in AUC with the addition of the GNN branch. This confirms the effectiveness of the GNN branch in modeling pixel relationships, thus further improving target detection performance.

The ROC curves indicate that the curve of Model 4 is closest to the top-left corner, while the curves of Models 2 and 3 are positioned to the top-left of the curve of Model 1, further validating the conclusions above. The separability graphs demonstrate that Model 1 has the poorest separability, and Model 4 exhibits the best separability, further confirming the effectiveness of the proposed DSC module and GNN branch.

TABLE II
AUC VALUES OF TEN COMPARISON DETECTORS ON FOUR BASELINE DATASETS

AUC	HYDICE	San Diego	Urban-1	Airport-4
OSP	0.97805	0.93415	0.91611	0.98112
CEM	0.99582	0.99251	0.96686	0.98957
SAM	0.69908	0.98477	0.99369	0.96079
CSCR	0.95958	0.99525	0.98454	0.98682
SRTD	0.99942	0.99583	0.98286	0.99558
CNNTD	0.95191	0.99462	0.90231	0.90287
HTD-Net	0.88609	0.93723	0.95954	0.99332
TSCNTD	0.98505	0.99502	0.98761	0.99491
STTD	0.99949	0.99625	0.99311	0.99501
Proposed	**0.99973**	**0.99657**	**0.99627**	**0.99691**

F. Quantitative Analysis of Experimental Results

Fig. 9 and Table II depict the AUC values, ROC curves, and separability maps of the detection results for ten HTD algorithms on four benchmark datasets. Overall, CEM outperforms SAM and OSP in terms of AUC values and ROC curves. However, on the Urban-1 dataset, SAM performs better than CEM, attributed to the high spectral resolution of this dataset, which encompasses rich spectral information. AUC values and ROC curves for OSP and SAM are unstable across the four datasets, indicating a lack of generalizability. Notably, SAM exhibits remarkably low AUC values on the HYDICE dataset due to its low spectral resolution and limited spectral

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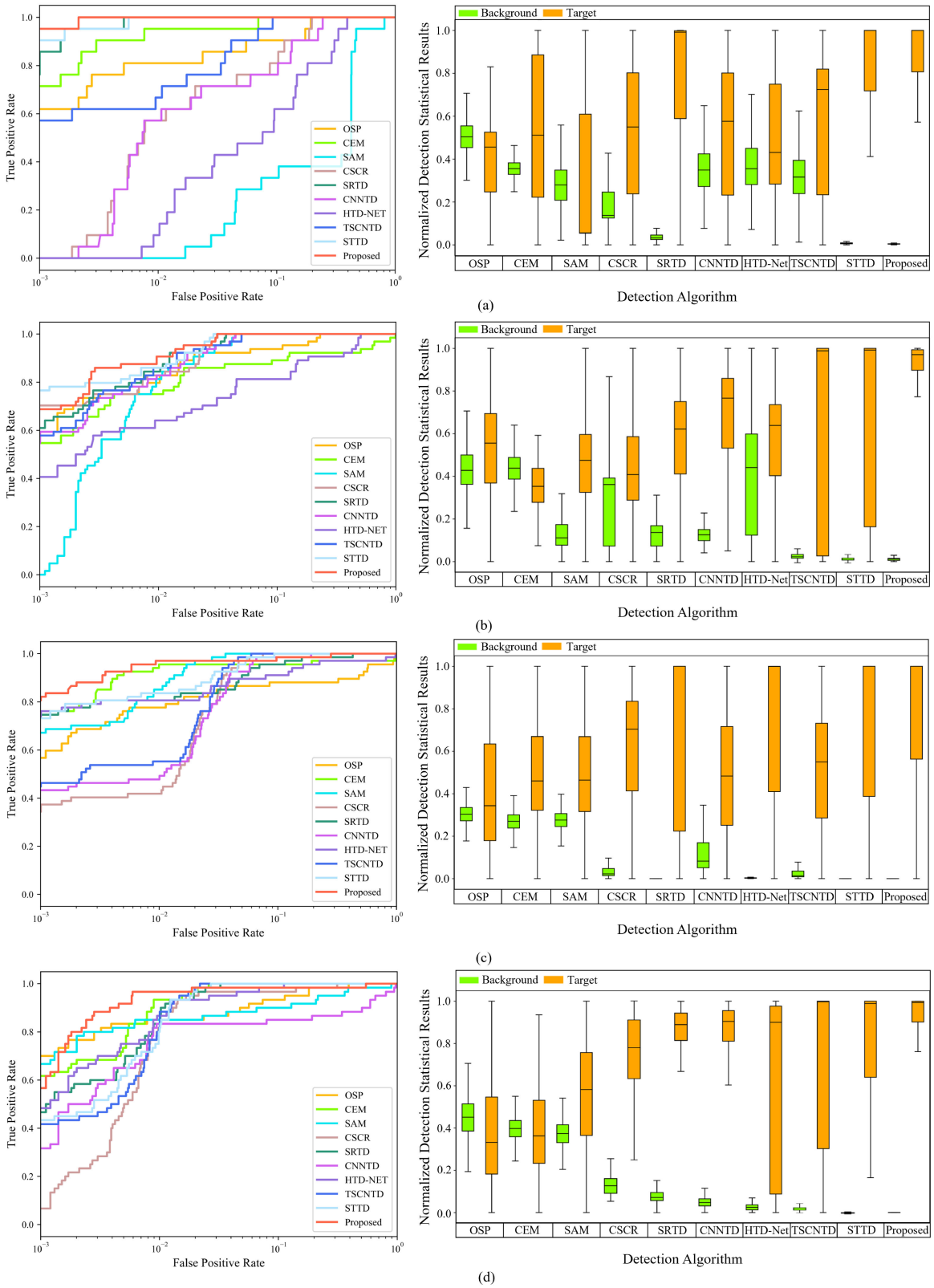


Fig. 9. (Left) ROC curves and (right) separability maps of ten comparison detectors on four baseline datasets. (a) HYDICE. (b) San Diego. (c) Urban-1. (d) Airport-4.

information. For most datasets, SAM's green box is closer to the coordinate 0, and farther from the orange box, indicating superior background suppression and separation capabilities. The two machine learning-based methods, reliant on expert knowledge, exhibit less stable AUC values. SRTD outperforms CSCR in AUC values and ROC curves across most datasets.

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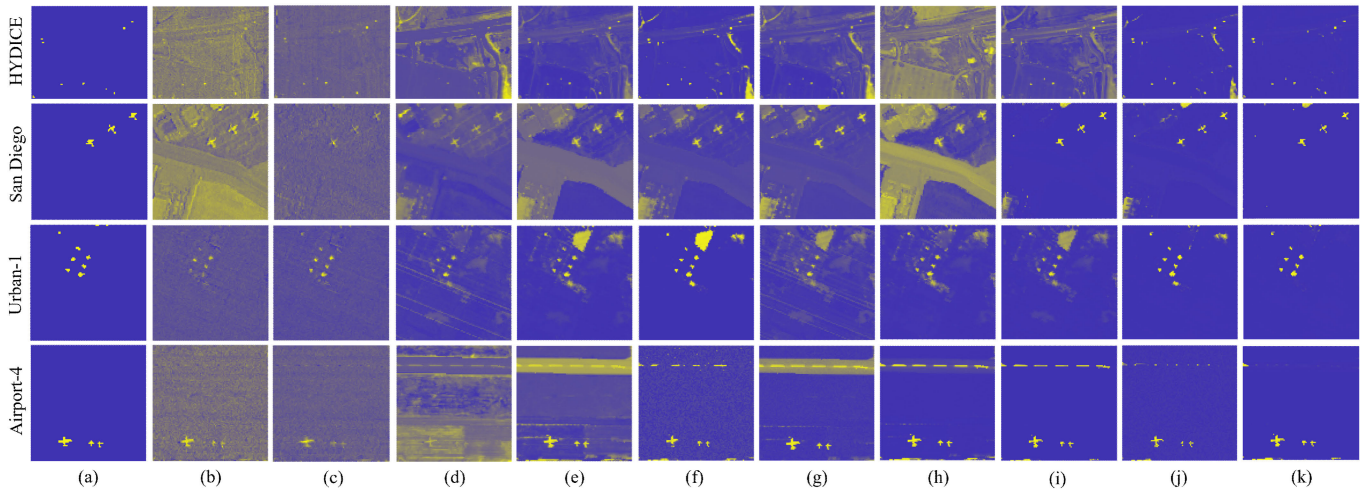


Fig. 10. Ground-truth maps and detection maps of ten comparison detectors on four baseline datasets. (a) Ground truth. (b) OSP. (c) CEM. (d) SAM. (e) CSCR. (f) SRTD. (g) CNNTD. (h) HTD-Net. (i) TSCNTD. (j) STTD. (k) Proposed.

Separability maps reveal that, across all datasets, SRTD's green box is closer to 0, the orange box is closer to 1, and the distance is greater, indicating superior capabilities in both background suppression and target highlighting compared to CSCR. In deep learning-based algorithms, CNNTD and HTD-Net show suboptimal and unstable performance. As both methods utilize pixelwise difference as input for training, they risk losing important feature information. Separability maps suggest that they exhibit good background suppression and target highlighting capabilities. TSCNTD and STTD achieve high levels of AUC values and ROC curves. Separability maps also indicate their strong background suppression capabilities. However, for some datasets, their separation between targets and background is not high. Notably, our proposed CFGC achieves the highest AUC values on all four datasets, indicating its excellent generalization. This is mainly attributed to modeling pixel relationships, integrating long-range spatial information with local spatial information, and fully utilizing spectral features, thereby minimizing the impact of dataset characteristics on model performance. From Fig. 9, it is evident that CFGC's red curve is closest to the upper-left corner, resulting in the largest AUC values. Separability maps further reveal that CFGC's green box is almost at 0, with a significant distance from the yellow box. In contrast, other detectors' green and orange boxes overlap or have smaller gaps, indicating that CFGC has superior background suppression capabilities.

The execution times of target detection methods based on deep learning are presented in Table III. It is evident from the table that as the image size increases and the spectral bands become more numerous, the required time also increases. Due to the use of large convolutional kernels and multiple convolutional layers, CNNTD incurs the longest training and testing time among the methods considered. The runtime of HTD exceeds that of TSCNTD, attributed to the larger number of training samples in HTD. STTD exhibits a longer runtime compared to HTD, attributed to the introduction of the transformer network, resulting in a more complex model and consequently longer training and testing time. Our proposed method, in comparison with other approaches, demonstrates the minimal training and testing times, thus saving substantial time.

TABLE III
EXECUTION TIME (IN SECONDS) COMPARISON OF DEEP LEARNING-BASED DETECTORS FOR FOUR DATASETS

Methods		HYDICE	San Diego	Urban	Airport
CNNTD	Train	1588.39	2581.33	2971.96	1764.81
	Test	99.65	113.05	152.05	101.54
HTD-Net	Train	875.48	920.49	923.78	893.63
	Test	7.62	8.13	8.54	7.81
TSCNTD	Train	643.02	692.13	704.25	698.09
	Test	4.67	**4.94**	5.89	6.75
STTD	Train	850.42	941.98	956.92	929.11
	Test	9.68	10.55	11.36	10.36
Proposed	Train	**165.48**	**205.08**	**210.32**	**229.45**
	Test	**4.52**	5.16	**5.47**	**5.69**

G. Visual Evaluation of Experimental Results

Fig. 10 illustrates the detection results of ten algorithms across four datasets. Among the three traditional target detection methods, CEM exhibits the most effective background suppression while simultaneously highlighting targets. OSP demonstrates the weakest background suppression performance. Additionally, while it almost distinctly outlines all targets, there is some confusion between targets and background, particularly noticeable in the HYDICE and San Diego datasets. SAM shows a relatively neutral stance between target highlighting and background suppression, yielding suboptimal visual separation results across datasets. SRTD and CSCR demonstrate pronounced accuracy in target pixel localization but exhibit weaker background suppression performance. Deep learning-based target detection algorithms effectively highlight target samples. Among them, CNNTD and HTD-Net exhibit limited capability in background suppression, leading to suboptimal visual separation results. TSCNTD and STTD perform well in background suppression, yet the visual separation

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of background remains cluttered, with some portions inadequately suppressed. Notably, the proposed CFGC algorithm exhibits superior visual separation effects. Across the four datasets, it not only effectively suppresses nearly all background but also highlights target samples.

V. Conclusion

In our study, we innovatively integrate GNN with CNN for HTD, introducing a novel network named CFGC. Initially, we employ the SSC algorithm and the similarity measurement strategy to select representative background samples. Subsequently, we propose Lapmix for constructing target samples to address the imbalance of HSI. Lapmix introduces target diversity without inducing excessive bias. Finally, we present CFGC as a HTD network that emphasizes the target samples while suppressing the background samples. CFGC comprises both CNN and GNN branches. In the CNN branch, we first extract multiscale features and then utilize the DSC-SA module to enhance features while reducing CNN parameters. In the GNN branch, we employ the LAGAB module to extract the global information. Through an additive fusion strategy, we obtain the fused features from both branches. Extensive experiments on four datasets are conducted, comparing with nine state-of-the-art target detection algorithms, validating the robustness and superiority of the proposed CFGC model.

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